WORKED SOLUTION

STEP III 1988, Question 6

Pure • Repeated Differentiation

STEP III 1988, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q6

The key idea

Differentiating a product 1988 times directly is hopeless, but a product of sines and cosines can be rewritten as a *sum*. Each term $\sin kx$ then differentiates in a four-step cycle, picking up a factor k each time. For the zero, the huge factor 3^{1988} in front of $\sin 3x$ means the derivative can only vanish where $\sin 3x$ is almost zero, so we look just past the first zero of $\sin 3x$ and linearise.

Solution

The 1988th derivative

By the product-to-sum formula $2\sin A\cos B = \sin(A+B)+\sin(A-B)$,

$$\mathrm{f}(x) = \sin 2x\cos x = \tfrac{1}{2}\left(\sin 3x + \sin x\right).$$

Differentiating $\sin kx$ four times gives

$$\sin kx \;\to\; k\cos kx \;\to\; -k^2\sin kx \;\to\; -k^3\cos kx \;\to\; k^4\sin kx,$$

so every four differentiations simply multiply $\sin kx$ by k^4. (Equivalently, the nth derivative of $\sin kx$ is $k^n\sin\left(kx+\tfrac{1}{2}n\pi\right)$.) Since $1988 = 4\times 497$,

$$\mathrm{f}^{(1988)}(x) = \tfrac{1}{2}\left(3^{1988}\sin 3x + \sin x\right).$$

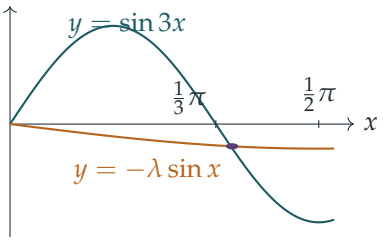
The smallest positive zero

We need the smallest $x>0$ with

$$3^{1988}\sin 3x + \sin x = 0. \qquad (*)$$

No zero in $(0,\tfrac{1}{3}\pi]$. For $0<x<\tfrac{1}{3}\pi$ we have $0<3x<\pi$, so both $\sin 3x$ and $\sin x$ are positive and the left-hand side of $(*)$ is positive. At $x=\tfrac{1}{3}\pi$ it equals $0+\tfrac{\sqrt{3}}{2}>0$.

A zero just after $\frac{1}{3}\pi$. At $x = \frac{1}{2}\pi$ the left-hand side is $3^{1988}\sin\frac{3}{2}\pi + 1 = 1 - 3^{1988} < 0$. It is continuous, so by the intermediate value theorem there is a zero in $(\frac{1}{3}\pi, \frac{1}{2}\pi)$, and the smallest positive zero is the first of these. Write it as $x = \frac{1}{3}\pi + \epsilon$ with $0 < \epsilon < \frac{1}{6}\pi$.



The sketch shows $(*)$ as $\sin 3x = -\lambda \sin x$ with $\lambda = 3^{-1988}$ (drawn far larger than it really is). The curves first meet just after $\sin 3x$ crosses zero at $\frac{1}{3}\pi$; for the true, tiny λ the crossing is pushed almost onto $\frac{1}{3}\pi$.

Since $\sin(\pi + 3\epsilon) = -\sin 3\epsilon$, equation $(*)$ becomes

$$3^{1988}\sin 3\epsilon = \sin\left(\tfrac{1}{3}\pi + \epsilon\right). \qquad (**)$$

Ask yourself How do we know ϵ is small enough to use $\sin 3\epsilon \approx 3\epsilon$?

The right-hand side of $(**)$ is at most 1, so $\sin 3\epsilon \le 3^{-1988}$. As $0 < 3\epsilon < \frac{1}{2}\pi$, and $\sin t \ge \frac{2t}{\pi}$ on $[0, \frac{1}{2}\pi]$ (the sine curve lies above its chord there), we get $3\epsilon \le \frac{\pi}{2}3^{-1988}$. So ϵ is astronomically small, which justifies the approximations below.

With ϵ this small, $\sin 3\epsilon = 3\epsilon\,(1 + O(\epsilon^2))$ and $\sin(\frac{1}{3}\pi + \epsilon) = \frac{\sqrt{3}}{2} + O(\epsilon)$. Substituting into $(**)$,

$$3^{1989}\,\epsilon \approx \frac{\sqrt{3}}{2}, \qquad \text{so} \qquad \epsilon \approx \frac{3^{-1989}\sqrt{3}}{2} = \frac{1}{3}\cdot\frac{3^{-1988}\sqrt{3}}{2},$$

with relative error of order 3^{-1988}.

A word about the printed value. The paper asks for $\epsilon \approx \frac{1}{2}3^{-1988}\sqrt{3}$, which is three times the value found here. The factor 3 comes from $\sin 3\epsilon \approx 3\epsilon$, and the numerical checks below confirm 3^{-1989}, not 3^{-1988}. So the question appears to contain a slip: the method it intends is exactly the one above, and only the power of 3 is off by one. If you reached 3^{-1989}, trust your working.

Checks

- The same argument works for any multiple n of 4: the first positive zero of $3^n\sin 3x + \sin x$ is at $\frac{1}{3}\pi + \epsilon_n$ with $\epsilon_n \approx \frac{1}{2}3^{-(n+1)}\sqrt{3}$. For $n = 4$, solving numerically gives $\epsilon_4 = 3.5713 \times 10^{-3}$, against the prediction $\frac{1}{2}\sqrt{3}/3^5 = 3.5640 \times 10^{-3}$ (ratio 1.0021). For $n = 8$ the ratio is 1.000025, and for $n = 40$ and beyond it is 1 to machine precision. The ratio to $\frac{1}{2}\sqrt{3}/3^n$ tends to $\frac{1}{3}$.

- The derivative formula can be checked on the first few cases: $\mathrm{f}^{(4)}(x) = \frac{1}{2}(81\sin 3x + \sin x)$, which you can confirm by differentiating $\frac{1}{2}(\sin 3x + \sin x)$ four times.

Common slips

- Getting the sign or phase wrong: only after a multiple of four differentiations does $\sin kx$ return to $+k^n \sin kx$.
- Forgetting to rule out zeros in $(0, \frac{1}{3}\pi)$; the sign argument there is what makes $\frac{1}{3}\pi + \epsilon$ the *smallest* positive zero.
- Writing $\sin(\pi + 3\epsilon) \approx \epsilon$ instead of -3ϵ, losing either the sign or the factor 3.

Beyond STEP

The second part is a simple instance of *perturbation theory*: a large parameter forces the solution to sit very close to a known root, and linearising about that root gives the leading correction, with the size of the error controlled by the next term in the Taylor series. Making "approximately equal" precise with $O(\cdot)$ notation and error bounds is a central theme of a first analysis course; see Abbott, *Understanding Analysis*.

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